Conditional Deterministic Dimension Amplification from Confident Feature Maps

July 22, 2026

1 Theoretical Setup

1.1 Problem setting

Fix a tie label $\tau \in \{-1,+1\}$, equal to the convention used for sign(0) in the learning protocol, and define

$$\operatorname{sgn}_\tau(z) := \begin{cases} +1, & z > 0, \\ -1, & z < 0, \\ \tau, & z = 0. \end{cases}$$

Let $\mathcal{X} = \{-1,+1\}^n$, let $\Delta(\mathcal{X})$ denote the set of probability distributions on $\mathcal{X}$, and let $\mathcal{H} \subseteq \{-1,+1\}^{\mathcal{X}}$. Since $\mathcal{X}$ is finite, $\mathcal{H}$ is finite. Its deterministic dimension complexity is

$$\operatorname{dc}(\mathcal{H}) := \min \left\{ q \in \mathbb{Z}_{\geq 0} : \begin{array}{l} \exists \Phi : \mathcal{X} \to \mathbb{R}^q \ \forall h \in \mathcal{H} \ \exists u_h \in \mathbb{R}^q \ \forall x \in \mathcal{X}, \\ \operatorname{sgn}_\tau(\langle u_h, \Phi(x) \rangle) = h(x) \end{array} \right\}.$$

We set $\operatorname{dc}(\varnothing) = 0$ and use $\mathbb{R}^0 = \{0\}$ and $\langle 0, 0 \rangle = 0$.

Fix a fully connected, bias-free ReLU architecture with $n_0 = n$, $n_L = 1$, positive integer widths $n_1, \ldots, n_{L-1}$, parameter matrices $\theta_i \in \mathbb{R}^{n_i \times n_{i-1}}$, and parameter count

$$S := \sum_{i=1}^{L} n_i n_{i-1}.$$

For $z_0 = x$, set $z_i = \sigma(\theta_i z_{i-1})$ for $1 \leq i < L$, where $\sigma(a) = \max\{0, a\}$ coordinatewise, and set $f_\theta(x) = \theta_L z_{L-1}$. Initialize independently according to

$$(\theta_i^{(0)})_{ab} \sim \mathcal{N}(0, 1/n_{i-1}).$$

At ReLU kinks, ∇^{src} denotes the fixed gradient or subgradient convention of the protocol; it makes the recursion single-valued without assuming that kinks are avoided. Given $\mathcal{D} \in \Delta(\mathcal{X})$ and $h^\star \in \mathcal{H}$, draw $x^{(0)}, \ldots, x^{(T-1)} \overset{\text{iid}}{\sim} \mathcal{D}$ and perform

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta^{\mathrm{src}} \ell\Big(h^\star(x^{(t)}) f_{\theta^{(t)}}(x^{(t)})\Big), \qquad \ell(z) = \log(1 + e^{-z}), \quad 0 \leq t < T.$$

The latter-half aggregate and returned classifier are

$$A_{\mathcal{D},h^\star}(x) := \sum_{t=\lceil T/2 \rceil}^{T} f_{\theta^{(t)}}(x), \qquad \widehat{h}_{\mathcal{D},h^\star}(x) := \operatorname{sgn}_\tau(A_{\mathcal{D},h^\star}(x)).$$

For any binary classifier g, write

$$\mathcal{L}_{\mathcal{D},h}(g) := \Pr_{x \sim \mathcal{D}}[g(x)h(x) < 0] = \Pr_{x \sim \mathcal{D}}[g(x) \neq h(x)].$$

1.2 Assumptions

Assumption 1 (Source parameter regime). *The integers n, L, T are positive, all network widths are positive integers with $n_0 = n$ and $n_L = 1$, $\eta > 0$, $0 \le \varepsilon < 1/4$, and the confident-map dimension $d \in \mathbb{Z}_{\ge 0}$ is an exposed parameter.*

Assumption 2 (Universal exact-SGD success). *The architecture, η, and T are fixed before the distribution and target are chosen, and*

$$\forall \mathcal{D} \in \Delta(\mathcal{X})\ \forall h^\star \in \mathcal{H}, \qquad \mathbb{E}_{\theta^{(0)},\, x^{(0:T-1)} \overset{\text{iid}}{\sim} \mathcal{D}} \left[\mathcal{L}_{\mathcal{D},h^\star}(\widehat{h}_{\mathcal{D},h^\star}) \right] \le \varepsilon.$$

The expectation is over exactly the Gaussian initialization and the one-sample SGD draws described above; the loss, update, architecture, and returned predictor are unchanged.

Assumption 3 (Target-independent tie-resolved confident map). *There exists one probability law $\mathcal{P}$ on maps $\phi : \mathcal{X} \to \mathbb{R}^d$, selected before $\mathcal{D}$ and h, such that*

$$\exists \mathcal{P}\ \forall \mathcal{D} \in \Delta(\mathcal{X})\ \forall h \in \mathcal{H}, \qquad \Pr_{\phi \sim \mathcal{P}} \Big[\exists w \in \mathbb{R}^d\ \forall x \in \mathcal{X},\ \operatorname{sgn}_\tau(\langle w, \phi(x) \rangle) = h(x) \Big] \ge \frac{1}{2}.$$

Thus $\mathcal{P}$ is independent of the subsequently quantified distribution and target, whereas w may depend on (ϕ, h). The event is exact and tie-resolved; no nonzero margin or avoidance of zero scores is assumed.

1.3 Formalized goal

The objective is the following conditional deterministic amplification statement with the explicit universal constant 7:

$$\operatorname{dc}(\mathcal{H}) \le 7TSd.$$

The constant must have no dependence on $n, \mathcal{H}, L, (n_i)_{i=0}^L, S, \eta, T, \varepsilon, d$, or $\mathcal{P}$. The proof must derive $\mathrm{VC}(\mathcal{H}) < 2T$, the finite-class count, and one deterministic common feature map. If a separate result also establishes an explicit polynomial bound $d \le p(S,T)$ with no hidden n- or η-dependence, the same theorem specializes to $\operatorname{dc}(\mathcal{H}) \le 7TSp(S,T)$. Assumption 3, and in particular such a polynomial bound on d, is an additional conditional premise; it is not claimed to follow from Assumption 2.

2 Preliminaries

A finite set $Z \subseteq \mathcal{X}$ is *shattered* by $\mathcal{H}$ if every map $b : Z \to \{-1, +1\}$ is the restriction of some member of $\mathcal{H}$. The VC dimension $\mathrm{VC}(\mathcal{H})$ is the largest cardinality of a shattered subset of $\mathcal{X}$. This setting-derived quantity is introduced only to eliminate the class cardinality in the proof of the main theorem; it does not remain in the final bound.

All feature representations use the Euclidean inner product and the same fixed map sgn_τ. In particular, equality of two scores preserves their predicted labels even when both scores are zero. This convention is used in the direct-sum construction proving the main theorem.

3 Main Theorem

Theorem 3.1 (Conditional deterministic dimension amplification). *Under Assumptions 1, 2, and 3,*

$$\operatorname{dc}(\mathcal{H}) \le 7TSd.$$

The conclusion is deterministic and gives exact tie-resolved representation on every point of $\mathcal{X}$. It is nonasymptotic and uses the fixed finite horizon T. The exposed variables in the rate are S, T, d; there is no hidden constant, and the numerical factor 7 is independent of $n, \mathcal{H}, L, (n_i)_{i=0}^{L}, S, \eta, T, \varepsilon, d$, and $\mathcal{P}$. The probability and expectation in the assumptions are used only to prove deterministic structural and existence statements; no confidence parameter, approximation error, margin, or norm change occurs in the conclusion.

This result is conditional on the target-independent confident-map premise. It does not assert that exact SGD success by itself produces that premise or a polynomial value of d.

Corollary 3.1 (Polynomial specialization under a separate dimension bound)**.** *Under the assumptions of Theorem 3.1, suppose a separate result establishes an explicit polynomial expression $p(S,T)$ satisfying $d \leq p(S,T)$ at the current parameters, with all coefficients and auxiliary dependence exposed and with no hidden dependence on n or η. Then*

$$\mathrm{dc}(\mathcal{H}) \leq 7TSp(S,T).$$

This is a deterministic, fixed-horizon, exact tie-resolved specialization and does not claim that the additional inequality follows from SGD.

Proof. Theorem 3.1 gives $\mathrm{dc}(\mathcal{H}) \leq 7TSd$. Assumption 1 and the positive widths give $T, S \geq 1$, while $d \leq p(S,T)$ gives $p(S,T) - d \geq 0$. Therefore

$$7TSp(S,T) - 7TSd = 7TS\big(p(S,T) - d\big) \geq 0,$$

and the displayed conclusion follows. This scalar substitution changes no probability mode, horizon, representation map, score, or tie convention. If $d = 0$, Theorem 3.1 itself forces exact dimension zero; if $p(S,T) = d$, the bridge is equality. $\square$

4 Proof Sketch

The empty class and the zero-dimensional case are first closed exactly. In the remaining case, the first parameter block gives $S \geq n$; see Proposition A.2.

To control the size of the class, suppose that $2T$ points are shattered and place the uniform distribution on them. Give those points independent fair labels and run the exact learner on the corresponding target. Conditional on the initialization, ordered sampled inputs, and sampled labels, an unsampled test label remains fair. The test point is unsampled with probability $(1-1/(2T))^T \geq 1/2$, so the average expected error over the finitely many realized targets is at least $1/4$. Selecting one fixed target contradicts $\varepsilon < 1/4$, proving Proposition A.4.

A self-contained finite-domain growth argument then bounds $|\mathcal{H}|$. Treating VC dimension zero separately and paying every ceiling explicitly gives, for $r = \lceil \log_2(2|\mathcal{H}|) \rceil$,

$$r \leq 7TS;$$

this is Proposition A.5. Draw r independent maps from the single target-independent law in Assumption 3. A fixed target is missed by every block with probability at most 2^{-r}, and the union bound over the finite class is at most $|\mathcal{H}|2^{-r} \leq 1/2$. Hence one deterministic tuple covers every target, as stated in Proposition A.6.

Finally, concatenate the tuple. For each target, place one successful separator in its successful block and zeros elsewhere. Lemma A.8 proves exact pointwise score equality with that block, including at ties. Thus a single rd-dimensional feature map represents all targets, and Proposition A.8 gives $\mathrm{dc}(\mathcal{H}) \leq rd \leq 7TSd$.

A Proof Details

A.1 Boundary cases and the architecture count

Proposition A.1 (Exact empty-class closure). *Under Assumption 1 and the convention* $\mathrm{dc}(\varnothing) = 0$*, if* $\mathcal{H} = \varnothing$*, then*

$$\mathrm{dc}(\mathcal{H}) = 0 \le 7TSd.$$

Proof. The defining convention gives $\mathrm{dc}(\mathcal{H}) = \mathrm{dc}(\varnothing) = 0$. Assumption 1 gives $T \ge 1$, $d \ge 0$, and a nonempty sum $S = \sum_{i=1}^{L} n_i n_{i-1}$ of positive integer products. Hence $7TSd \ge 0$, which proves the inequality. □

Lemma A.1 (Zero-dimensional tie-resolved closure). *Under Assumptions 1 and 3, and the conventions* $\mathbb{R}^0 = \{0\}$*,* $\langle 0, 0\rangle = 0$*, and* $\mathrm{sgn}_\tau(0) = \tau$*, if* $\mathcal{H} \ne \varnothing$ *and* $d = 0$*, then every* $h \in \mathcal{H}$ *satisfies* $h(x) = \tau$ *for all* $x \in \mathcal{X}$*, and*

$$\mathrm{dc}(\mathcal{H}) = 0 = 7TSd.$$

Proof. Since $\mathbb{R}^0$ is the singleton $\{0\}$, there is exactly one map from $\mathcal{X}$ to $\mathbb{R}^0$, namely ϕ_0 with $\phi_0(x) = 0$ for every x. Therefore every probability law on such maps, including the law $\mathcal{P}$ in Assumption 3, assigns probability one to ϕ_0. There is also exactly one candidate separator $w \in \mathbb{R}^0$, namely $w = 0$.

Fix any $h \in \mathcal{H}$. Because $n \ge 1$, the domain $\mathcal{X} = \{-1, +1\}^n$ is nonempty, so the confident-map premise may be instantiated at any distribution on $\mathcal{X}$. Its success event for this fixed h is

$$E_h := \left\{\phi : \ \exists w \in \mathbb{R}^0 \ \forall x \in \mathcal{X}, \mathrm{sgn}_\tau(\langle w, \phi(x)\rangle) = h(x)\right\}.$$

For the sole possible map ϕ_0 and the sole separator,

$$\begin{aligned} \phi_0 \in E_h &\iff \forall x \in \mathcal{X}, \mathrm{sgn}_\tau(\langle 0, 0\rangle) = h(x) \\ &\iff \forall x \in \mathcal{X},\ h(x) = \tau. \end{aligned}$$

Thus E_h is deterministic: its $\mathcal{P}$-probability is either zero or one. Assumption 3 gives $\mathcal{P}(E_h) \ge 1/2$, so $\mathcal{P}(E_h) = 1$ and $h(x) = \tau$ for every x. Since h was arbitrary, every member of $\mathcal{H}$ is the constant τ classifier.

The same map ϕ_0, together with the separator $u_h = 0$ for every $h \in \mathcal{H}$, therefore satisfies

$$\mathrm{sgn}_\tau(\langle u_h, \phi_0(x)\rangle) = \mathrm{sgn}_\tau(0) = \tau = h(x) \qquad \text{for all } h \in \mathcal{H},\ x \in \mathcal{X}.$$

Dimension $q = 0$ is consequently feasible in the definition of $\mathrm{dc}(\mathcal{H})$. Since that definition minimizes over $q \in \mathbb{Z}_{\ge 0}$, it follows that $\mathrm{dc}(\mathcal{H}) = 0$. Finally, $d = 0$ gives $7TSd = 0$, proving exact equality. □

Proposition A.2 (First-layer structural bound). *Under Assumption 1, if* $\mathcal{H} \ne \varnothing$ *and* $d \ne 0$*, then*

$$d \ge 1, \qquad S \ge n \ge 1, \qquad T \ge 1, \qquad S \ge 1.$$

Moreover, if $L = 1$*, then* $S = n$*.*

Proof. Assumption 1 states that $d \in \mathbb{Z}_{\ge 0}$. Hence the branch condition $d \ne 0$ implies $d \ge 1$. The same assumption gives $n, L, T \in \mathbb{Z}_{\ge 1}$, so $n \ge 1$ and $T \ge 1$.

For every allowed depth, $n_1 \ge 1$: when $L \ge 2$, it is a positive hidden width, while when $L = 1$, it is the output width $n_1 = n_L = 1$. The first summand in the definition of S therefore obeys

$$n_1 n_0 = n_1 n \ge n.$$

All summands in $S = \sum_{i=1}^{L} n_i n_{i-1}$ are positive, so

$$S \geq n_1 n_0 \geq n \geq 1.$$

This proves both $S \geq n$ and $S \geq 1$. If $L = 1$, the sum has only its first term and $n_1 = n_L = 1$; hence

$$S = n_1 n_0 = 1 \cdot n = n.$$

Thus the same count covers the depth-one boundary without presupposing a hidden layer. $\square$

Boundary reduction. The case split is exhaustive. If $\mathcal{H} = \varnothing$, Proposition A.1 proves the target exactly. Otherwise $\mathcal{H} \neq \varnothing$. In that branch, if $d = 0$, Lemma A.1 proves $\mathrm{dc}(\mathcal{H}) = 0 = 7TSd$ with the fixed tie convention. If instead $d \neq 0$, Proposition A.2 gives

$$\mathcal{H} \neq \varnothing, \qquad d \geq 1, \qquad S \geq n \geq 1, \qquad T, S \geq 1,$$

and explicitly includes the $L = 1$ case. These are exactly the boundary conclusions and remaining-branch structural controls needed below. $\square$

A.2 A VC ceiling from the exact learner

Lemma A.2 (Sampled-label measurability of exact SGD)**.** *Under Assumption 1, suppose $Z \subseteq \mathcal{X}$ has $m = 2T$ points and, for every $b \in \{-1, +1\}^Z$, a fixed target $h_b \in \mathcal{H}$ satisfies $h_b(z) = b(z)$ for every $z \in Z$. Let*

$$B = (B_z)_{z \in Z}, \qquad B_z \overset{\text{iid}}{\sim} \mathrm{Unif}\{-1, +1\},$$

let $X_0, \ldots, X_{T-1}$ be iid uniform on Z, let X be an independent uniform point of Z, and draw the source Gaussian initialization $\Theta^{(0)}$ independently of $B, X_0, \ldots, X_{T-1}, X$. Run the exact source recursion with target h_B. Then, for every $0 \leq t \leq T$,

$$\Theta^{(t)} \textit{ is measurable with respect to } \mathscr{F}_t := \sigma\Big(\Theta^{(0)}, (X_s, B_{X_s})_{0 \leq s < t}\Big),$$

and the returned prediction at X,

$$\widehat{g}_B(X) := \widehat{h}_{\mathcal{D}_Z, h_B}(X), \qquad \mathcal{D}_Z := \mathrm{Unif}(Z),$$

is measurable with respect to

$$\mathscr{F} := \sigma\Big(X, \Theta^{(0)}, (X_s, B_{X_s})_{s=0}^{T-1}\Big).$$

Proof. For a parameter value θ, an input $x \in \mathcal{X}$, and a label $y \in \{-1, +1\}$, define the exact update map

$$U(\theta, x, y) := \theta - eta \nabla_\theta^{\mathrm{src}} \ell(y f_\theta(x)).$$

All conventions in this expression are fixed before the target is selected. In particular, the fixed source choice at every ReLU kink makes U single-valued; it cannot encode an unsampled value of the target.

Because every X_t lies in Z, the target label passed to update t is exactly $h_B(X_t) = B_{X_t}$. The recursion therefore has the pathwise form

$$\Theta^{(t+1)} = U\Big(\Theta^{(t)}, X_t, B_{X_t}\Big).$$

At $t = 0$, $\Theta^{(0)}$ is $\mathscr{F}_0$-measurable. If $\Theta^{(t)}$ is $\mathscr{F}_t$-measurable, the displayed update shows that $\Theta^{(t+1)}$ is measurable with respect to

$$\sigma(\mathscr{F}_t, X_t, B_{X_t}) = \mathscr{F}_{t+1}.$$

Induction proves the iterate claim for all $0 \le t \le T$. This induction is pathwise and permits arbitrary repetitions among the X_t's.

Consequently every term $f_{\Theta^{(t)}}(X)$ in

$$\sum_{t=\lceil T/2 \rceil}^{T} f_{\Theta^{(t)}}(X)$$

is $\mathscr{F}$-measurable. Applying the fixed map sgn_τ yields an $\mathscr{F}$-measurable value in $\{-1, +1\}$, even when the aggregate is zero, because that case returns the fixed label τ. Thus $\widehat{g}_B(X)$ is determined by exactly the information listed in $\mathscr{F}$, and no unsampled coordinate of B enters the recursion or the output. □

Lemma A.3 (Fair unseen bit)**.** *Under Assumption 1 and the experiment and notation of Lemma A.2, define*

$$E := \{X \notin \{X_0, \ldots, X_{T-1}\}\}.$$

Then, for each $s \in \{-1, +1\}$,

$$\Pr(B_X = s \mid \mathscr{F}) = \frac{1}{2} \quad \textit{almost surely on } E,$$

and

$$\Pr(\widehat{g}_B(X) \ne B_X \mid \mathscr{F}) = \frac{1}{2} \quad \textit{almost surely on } E.$$

Proof. Condition first on concrete values

$$X = x, \qquad (X_0, \ldots, X_{T-1}) = (x_0, \ldots, x_{T-1}).$$

On E, x differs from every x_t. Let $I = \{x_0, \ldots, x_{T-1}\}$; repetitions merely make $|I| < T$ and do not add a new label coordinate. The observed label vector $(B_{x_t})_{t=0}^{T-1}$ is measurable with respect to $(B_z)_{z \in I}$. Since the coordinates $(B_z)_{z \in Z}$ are mutually independent fair signs and $x \notin I$, the coordinate B_x remains a fair sign after conditioning on every observed label. The initialization $\Theta^{(0)}$ was drawn independently of the complete label vector, so conditioning on it does not change this conclusion. Averaging this pointwise conditional statement over the conditioned input values gives

$$\Pr(B_X = s \mid \mathscr{F}) = \frac{1}{2} \quad \text{on } E.$$

The event E itself is $\mathscr{F}$-measurable because it depends only on $X, X_0, \ldots, X_{T-1}$.

By Lemma A.2, $\widehat{g}_B(X)$ is $\mathscr{F}$-measurable and takes one fixed value in $\{-1, +1\}$ under this conditioning. Therefore, on E,

$$\begin{aligned} \Pr(\widehat{g}_B(X) \ne B_X \mid \mathscr{F}) &= \sum_{s \in \{-1,+1\}} \mathbf{1}\{\widehat{g}_B(X) \ne s\} \Pr(B_X = s \mid \mathscr{F}) \\ &= \frac{1}{2}. \end{aligned}$$

If the aggregate score is zero, the first factor still contains the fixed binary value τ, so the same calculation applies without a no-tie or margin assumption. □

Lemma A.4 (Finite-horizon avoidance). *Under Assumption 1, if $X, X_0, \ldots, X_{T-1}$ are independent and uniform on a set Z of size $2T$, then*

$$\Pr[X \notin \{X_0, \ldots, X_{T-1}\}] = \left(1 - \frac{1}{2T}\right)^T \geq \frac{1}{2}.$$

For $T = 1$, both the exact probability and the displayed lower bound equal $1/2$.

Proof. Conditional on any fixed $X = x \in Z$, each X_t avoids x with probability

$$1 - \frac{1}{|Z|} = 1 - \frac{1}{2T}.$$

The T training draws are independent, including when their realized values repeat, so multiplication gives

$$\Pr(E \mid X = x) = \left(1 - \frac{1}{2T}\right)^T.$$

This expression does not depend on x, proving the equality after averaging over X.

For completeness, for every integer $k \geq 1$ and $u \in [0, 1]$,

$$(1 - u)^k \geq 1 - ku.$$

Indeed, equality holds for $k = 1$; if it holds for k, multiplication by $1 - u \geq 0$ gives

$$(1 - u)^{k+1} \geq (1 - ku)(1 - u) = 1 - (k + 1)u + ku^2 \geq 1 - (k + 1)u.$$

Applying this induction with $k = T$ and $u = 1/(2T) \in (0, 1]$ yields

$$\left(1 - \frac{1}{2T}\right)^T \geq 1 - \frac{T}{2T} = \frac{1}{2}.$$

When $T = 1$, the exact product is $1 - 1/2 = 1/2$, so the boundary case incurs no slack. $\square$

Proposition A.3 (Random-label average-risk lower bound). *Under Assumption 1 and Lemmas A.2, A.3, and A.4, if $Z \subseteq \mathcal{X}$ has $2T$ points and every labeling $b \in \{-1, +1\}^Z$ has a fixed representative $h_b \in \mathcal{H}$ with $h_b|_Z = b$, then, for $\mathcal{D}_Z = \mathrm{Unif}(Z)$,*

$$2^{-2T} \sum_{b \in \{-1,+1\}^Z} \mathbb{E}_{\Theta^{(0)}, X_{0:T-1} \overset{\text{iid}}{\sim} \mathcal{D}_Z} \left[\mathcal{L}_{\mathcal{D}_Z, h_b}(\widehat{h}_{\mathcal{D}_Z, h_b})\right] \geq \frac{1}{4}.$$

Proof. In the joint experiment of Lemma A.2, condition on $B = b$, the initialization, and the ordered training sample. The independent point $X \sim \mathcal{D}_Z$ converts the population risk into an indicator expectation. Since $X \in Z$ and $h_b|_Z = b$,

$$\mathcal{L}_{\mathcal{D}_Z, h_b}(\widehat{h}_{\mathcal{D}_Z, h_b}) = \mathbb{E}_X \left[\mathbf{1}\{\widehat{h}_{\mathcal{D}_Z, h_b}(X) \neq b(X)\} \,\middle|\, b, \Theta^{(0)}, X_{0:T-1}\right].$$

Taking expectation over the uniform labeling B, initialization, and training sample, and using that B has exactly 2^{2T} equiprobable values, gives the exact identity

$$\begin{aligned} &2^{-2T} \sum_{b \in \{-1,+1\}^Z} \mathbb{E}_{\Theta^{(0)}, X_{0:T-1}} \left[\mathcal{L}_{\mathcal{D}_Z, h_b}(\widehat{h}_{\mathcal{D}_Z, h_b})\right] \\ &\qquad = \Pr[\widehat{g}_B(X) \neq B_X]. \end{aligned}$$

Because $E \in \mathscr{F}$, the tower property and Lemma A.3 yield

$$\begin{aligned}\Pr[\widehat{g}_B(X) \neq B_X] &\geq \mathbb{E}[\mathbf{1}_E \mathbf{1}\{\widehat{g}_B(X) \neq B_X\}] \\ &= \mathbb{E}[\mathbf{1}_E \Pr(\widehat{g}_B(X) \neq B_X \mid \mathscr{F})] \\ &= \frac{1}{2}\Pr(E).\end{aligned}$$

Lemma A.4 gives $\Pr(E) \geq 1/2$, and hence the finite average is at least $1/4$. At $T = 1$, $\Pr(E) = 1/2$ exactly, so this argument still gives the exact lower bound $1/4$. □

Proposition A.4 (VC ceiling from universal exact-SGD success)**.** *Under Assumptions 1 and 2, and on the remaining branch identified by Proposition A.2,*

$$\mathrm{VC}(\mathcal{H}) < 2T.$$

Proof. Suppose instead that $\mathrm{VC}(\mathcal{H}) \geq 2T$. Then there is a set $Z \subseteq \mathcal{X}$ of exactly $2T$ points shattered by $\mathcal{H}$. For each of the finitely many $b \in \{-1, +1\}^Z$, shattering supplies at least one $h \in \mathcal{H}$ with $h|_Z = b$. Fix one such representative and call it h_b.

For each fixed labeling b, define

$$R_b := \mathbb{E}_{\Theta^{(0)}, X_{0:T-1} \overset{\mathrm{iid}}{\sim} \mathcal{D}_Z}\left[\mathcal{L}_{\mathcal{D}_Z, h_b}(\widehat{h}_{\mathcal{D}_Z, h_b})\right], \qquad \mathcal{D}_Z = \mathrm{Unif}(Z).$$

Proposition A.3 proves

$$2^{-2T} \sum_{b \in \{-1,+1\}^Z} R_b \geq \frac{1}{4}.$$

This is an average over a finite set, so at least one fixed labeling $b_\star$ satisfies $R_{b_\star} \geq 1/4$. The auxiliary random labeling used to establish the average is now gone: $\mathcal{D}_Z$ and $h_{b_\star} \in \mathcal{H}$ are fixed, and $R_{b_\star}$ averages only over the exact Gaussian initialization and the T iid training draws specified in the setting.

The architecture, step size η, and horizon T were fixed before this witness distribution and target were chosen. Therefore Assumption 2 applies directly to this same fixed pair $(\mathcal{D}_Z, h_{b_\star})$ and the same exact learner, giving

$$R_{b_\star} \leq \varepsilon < \frac{1}{4}.$$

This contradicts $R_{b_\star} \geq 1/4$. Thus no $2T$-point subset of $\mathcal{X}$ is shattered, and $\mathrm{VC}(\mathcal{H}) < 2T$. When $T = 1$, the lower bound remains $1/4$ and the strict assumption $\varepsilon < 1/4$ gives the same contradiction. □

VC-dimension conclusion. Proposition A.2 places the proof on the nondegenerate branch. Lemma A.2 shows from the exact update and fixed kink convention that the learner uses only sampled target labels. Lemma A.3 gives conditional error $1/2$ on an unsampled test point, including repeated inputs and zero aggregate scores. Lemma A.4 gives the exact probability $(1 - 1/(2T))^T \geq 1/2$, and Proposition A.3 converts these facts to the finite average risk lower bound. The finite selection and the strict comparison with $\varepsilon < 1/4$ in Proposition A.4 then yield the displayed VC ceiling. □

A.3 Finite-class counting and the repetition budget

Lemma A.5 (Finite-domain growth bound)**.** *Let A be a finite set of cardinality m, and let $\mathcal{G} \subseteq \{-1,+1\}^A$ be nonempty. If $q = \mathrm{VC}(\mathcal{G})$, then $0 \le q \le m$ and*

$$|\mathcal{G}| \le \sum_{j=0}^{q} \binom{m}{j}.$$

If $q = 0$, then in fact $|\mathcal{G}| = 1$.

Proof. A shattered set is a subset of A, so $0 \le q \le m$. We prove the cardinality inequality in the slightly more general form that every nonempty $\mathcal{G} \subseteq \{-1,+1\}^A$ satisfying $\mathrm{VC}(\mathcal{G}) \le q$, for an integer $0 \le q \le m$, obeys

$$|\mathcal{G}| \le \sum_{j=0}^{q} \binom{m}{j}. \tag{A.1}$$

Use induction on m, with the convention $\binom{a}{b} = 0$ when $b < 0$ or $b > a$.

If $m = 0$, there is exactly one function on $A = \varnothing$, so nonemptiness gives $|\mathcal{G}| = 1 = \binom{0}{0}$. For arbitrary m, if $q = 0$, two distinct functions in $\mathcal{G}$ would differ at some $a \in A$. Those two functions would realize both labels on $\{a\}$, thereby shattering that singleton and contradicting $\mathrm{VC}(\mathcal{G}) \le 0$. Hence $|\mathcal{G}| \le 1 = \binom{m}{0}$. When $\mathcal{G}$ is nonempty and its VC dimension is exactly zero, this also proves $|\mathcal{G}| = 1$. If $q = m$, then the trivial count of all binary functions gives

$$|\mathcal{G}| \le 2^m = \sum_{j=0}^{m} \binom{m}{j}.$$

It remains to prove (A.1) when $m \ge 1$ and $1 \le q < m$. Fix $a \in A$, put $B = A \setminus \{a\}$, and, for $s \in \{-1,+1\}$, define the restriction class

$$\mathcal{G}_s := \{g|_B : g \in \mathcal{G},\ g(a) = s\} \subseteq \{-1,+1\}^B.$$

Let

$$\mathcal{U} := \mathcal{G}_{+1} \cup \mathcal{G}_{-1}, \qquad \mathcal{P}_0 := \mathcal{G}_{+1} \cap \mathcal{G}_{-1}.$$

A function in $\mathcal{G}$ is uniquely determined by its restriction to B and its value at a. Inclusion–exclusion therefore gives the exact decomposition

$$|\mathcal{G}| = |\mathcal{G}_{+1}| + |\mathcal{G}_{-1}| = |\mathcal{U}| + |\mathcal{P}_0|. \tag{A.2}$$

If $\mathcal{U}$ shatters $D \subseteq B$, then each labeling of D has an extension in $\mathcal{G}$, so $\mathcal{G}$ also shatters D. Thus $\mathrm{VC}(\mathcal{U}) \le q$. If $\mathcal{P}_0$ is nonempty and shatters $D \subseteq B$, then for each labeling of D its realizing restriction in $\mathcal{P}_0$ has one extension in $\mathcal{G}$ with label $+1$ at a and another with label -1 there. Hence $\mathcal{G}$ shatters $D \cup \{a\}$, which shows $\mathrm{VC}(\mathcal{P}_0) \le q - 1$. If $\mathcal{P}_0$ is empty, the bound below is immediate for it.

Both $\mathcal{U}$ and $\mathcal{P}_0$ are classes on the $(m-1)$-point set B. Applying the induction hypothesis, and using the zero bound when $\mathcal{P}_0 = \varnothing$, gives

$$|\mathcal{U}| \le \sum_{j=0}^{q} \binom{m-1}{j}, \qquad |\mathcal{P}_0| \le \sum_{j=0}^{q-1} \binom{m-1}{j}.$$

For completeness, partitioning the j-element subsets of an m-element set according to whether they contain one fixed element gives

$$\binom{m}{j} = \binom{m-1}{j} + \binom{m-1}{j-1}.$$

Combining this identity with (A.2) yields

$$\begin{aligned}|\mathcal{G}| &\le \sum_{j=0}^{q} \binom{m-1}{j} + \sum_{j=0}^{q-1} \binom{m-1}{j} \\ &= \sum_{j=0}^{q} \left(\binom{m-1}{j} + \binom{m-1}{j-1} \right) = \sum_{j=0}^{q} \binom{m}{j}.\end{aligned}$$

This closes the induction and proves the lemma. □

Lemma A.6 (Binomial-sum and numerical logarithm estimates)**.** *For integers m, q satisfying $1 \le q \le m$,*

$$\sum_{j=0}^{q} \binom{m}{j} \le \left(\frac{em}{q}\right)^{q}.$$

Moreover,

$$\log_2 e < \frac{3}{2}.$$

Proof. Set $a = q/m \in (0, 1]$. For every $0 \le j \le q$, one has $a^j \ge a^q$. Expanding the product of m copies of $1 + a$, and grouping terms by the number j of copies from which a is selected, gives

$$(1 + a)^m = \sum_{j=0}^{m} \binom{m}{j} a^j.$$

All terms are nonnegative, and hence

$$\begin{aligned}\sum_{j=0}^{q} \binom{m}{j} &\le a^{-q} \sum_{j=0}^{q} \binom{m}{j} a^j \\ &\le a^{-q}(1 + a)^m.\end{aligned} \tag{A.3}$$

For $a \ge 0$, the exponential series gives

$$e^a = \sum_{k=0}^{\infty} \frac{a^k}{k!} \ge 1 + a.$$

Consequently $(1 + a)^m \le e^{am} = e^q$. Substituting this in (A.3) and using $a^{-q} = (m/q)^q$ proves

$$\sum_{j=0}^{q} \binom{m}{j} \le \left(\frac{em}{q}\right)^{q}.$$

It remains to verify the numerical logarithm bound without a decimal approximation. For every $k \ge 4$,

$$k! \ge 24\, 4^{k-4},$$

because $4! = 24$ and each subsequent factor is at least 4. Hence

$$e = \sum_{k=0}^{\infty} \frac{1}{k!} \leq 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \sum_{j=0}^{\infty} 4^{-j}$$
$$= 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{18} = \frac{49}{18}.$$

Since

$$\left(\frac{49}{18}\right)^2 = \frac{2401}{324} < \frac{2592}{324} = 8,$$

we have $e \leq 49/18 < \sqrt{8} = 2^{3/2}$. Monotonicity of the base-two logarithm now gives $\log_2 e < 3/2$. $\square$

Proposition A.5 (Explicit repetition budget)**.** *Under Assumption 1, Propositions A.2 and A.4, and Lemmas A.5 and A.6, on the nondegenerate branch define*

$$N := |\mathcal{X}| = 2^n, \qquad M := |\mathcal{H}|, \qquad v := \mathrm{VC}(\mathcal{H}), \qquad r := \lceil \log_2(2M) \rceil .$$

Then $r \leq 7TS$.

Proof. Proposition A.2 gives $\mathcal{H} \neq \varnothing$, so $M \geq 1$ and r is well-defined. It also gives

$$n \geq 1, \qquad S \geq n, \qquad T, S \geq 1. \tag{A.4}$$

Proposition A.4 gives

$$v < 2T. \tag{A.5}$$

First suppose $v = 0$. Applying Lemma A.5 to $A = \mathcal{X}$ and $\mathcal{G} = \mathcal{H}$ gives $M = 1$. Therefore

$$r = \lceil \log_2 2 \rceil = 1 \leq 7TS,$$

where the last inequality follows from $T, S \geq 1$. This handles the singular case in which $(eN/v)^v$ is unavailable.

Now suppose $v \geq 1$. Lemma A.5 gives $v \leq N$ and

$$M \leq \sum_{j=0}^{v} \binom{N}{j}.$$

Lemma A.6 therefore applies, including when $v = N$, and yields

$$M \leq \left(\frac{eN}{v}\right)^v.$$

Taking base-two logarithms, using $N = 2^n$ and $\log_2 v \geq 0$, and then using (A.5) and $\log_2 e < 3/2$, gives

$$\begin{aligned}\log_2 M &\leq v \log_2\left(\frac{eN}{v}\right)\\ &= v\,(n + \log_2 e - \log_2 v)\\ &\leq v\,(n + \log_2 e)\\ &\leq 2T\left(n + \frac{3}{2}\right)\\ &\leq 5Tn.\end{aligned} \tag{A.6}$$

The last inequality is explicit: $n \geq 1$ implies

$$n + \frac{3}{2} \leq \frac{5}{2}n.$$

For every real y, $\lceil y \rceil \leq y + 1$. Hence (A.6) accounts for the ceiling as

$$\begin{aligned} r &= \lceil 1 + \log_2 M \rceil \\ &\leq \log_2 M + 2 \\ &\leq 5Tn + 2 \\ &\leq 5TS + 2TS \\ &= 7TS. \end{aligned} \tag{A.7}$$

Here $5Tn \leq 5TS$ follows from $n \leq S$, while $2 \leq 2TS$ follows from $T, S \geq 1$. Thus neither n nor the additive ceiling contribution remains in the exported bound.

The corner $T = 1, n = 1$ is included explicitly: then $N = 2$ and (A.5) forces $v \in \{0, 1\}$, so one of the two branches above applies. In the $v \geq 1$ branch, $n + 3/2 = (5/2)n$; if also $S = 1$, then $2 = 2TS$. Thus the two potentially tight elementary absorptions in (A.6)– (A.7) remain valid at the smallest allowed values. ☐

Derivation of the repetition budget. Proposition A.2 supplies nonemptiness, $S \geq n \geq 1$, and $T, S \geq 1$, while Proposition A.4 supplies $v < 2T$. Lemma A.5 proves both the finite-domain growth bound and the exact $v = 0 \Rightarrow M = 1$ boundary. Lemma A.6 proves the binomial estimate and $\log_2 e < 3/2$. Proposition A.5 then pays the ceiling by $2 \leq 2TS$ and eliminates n using $n \leq S$, yielding

$$r = \lceil \log_2(2|\mathcal{H}|) \rceil \leq 7TS.$$

☐

A.4 From target-wise confidence to one common tuple

Lemma A.7 (Independent amplification for one target)**.** *Under Assumption 3 and Proposition A.5, fix the single law $\mathcal{P}$ supplied before all distributions and targets, and let*

$$r = \lceil \log_2(2|\mathcal{H}|) \rceil \geq 1.$$

If $\phi_1, \ldots, \phi_r$ are iid with law $\mathcal{P}$, then, for every fixed $h \in \mathcal{H}$,

$$\Pr_{(\phi_1,\ldots,\phi_r)\sim\mathcal{P}^r}\left[\begin{array}{c} \forall i \in \{1,\ldots,r\},\ \nexists w \in \mathbb{R}^d \textit{ such that} \\ \forall x \in \mathcal{X}, \quad \mathrm{sgn}_\tau(\langle w, \phi_i(x)\rangle) = h(x) \end{array}\right] \leq 2^{-r}.$$

Proof. Choose the one $\mathcal{P}$ occurring before the universal $\mathcal{D}$- and h-quantifiers in Assumption 3. For a fixed $h \in \mathcal{H}$, define the exact one-map event

$$E_h := \left\{\phi : \mathcal{X} \to \mathbb{R}^d : \ \exists w \in \mathbb{R}^d \ \forall x \in \mathcal{X}, \quad \mathrm{sgn}_\tau(\langle w, \phi(x)\rangle) = h(x)\right\}. \tag{A.8}$$

The probability in the assumption makes E_h measurable. The event contains no $\mathcal{D}$. The finite hypercube $\mathcal{X}$ is nonempty, so $\Delta(\mathcal{X})$ contains a point mass and the universal distribution quantifier is not vacuous. Consequently, after the single law $\mathcal{P}$ has been fixed, that quantifier does not change the event or permit a new law, and the premise gives

$$\mathcal{P}(E_h) \geq \frac{1}{2}, \qquad \mathcal{P}(E_h^c) \leq \frac{1}{2} \tag{A.9}$$

for every fixed $h \in \mathcal{H}$.

For coordinate i, the event that block i fails to represent h is $\{\phi_i \in E_h^c\}$. Since all coordinates have the same law $\mathcal{P}$ and are independent, the defining rectangle identity for the product law and (A.9) give

$$\begin{aligned} \Pr_{\mathcal{P}^r}\left[\bigcap_{i=1}^{r}\{\phi_i \in E_h^c\}\right] &= \prod_{i=1}^{r}\mathcal{P}(E_h^c) \\ &= \mathcal{P}(E_h^c)^r \\ &\le \left(\frac{1}{2}\right)^r = 2^{-r}. \end{aligned} \tag{A.10}$$

Membership in E_h is exactly the full-domain event in the statement, so (A.10) is the claimed failure bound. In particular, the calculation neither requires nor permits a law depending on h, and it never replaces sgn_τ by a strict-margin or approximate event. □

Proposition A.6 (Common deterministic exact-cover tuple)**.** *Under Assumption 3, Proposition A.5, and Lemma A.7, let*

$$M := |\mathcal{H}|, \qquad r := \lceil \log_2(2M) \rceil .$$

Then $1 \le r \le 7TS$, and there are deterministic maps $\phi_i^* : \mathcal{X} \to \mathbb{R}^d$, $1 \le i \le r$, such that

$$\forall h \in \mathcal{H}\ \exists i \in \{1,\dots,r\}\ \exists w \in \mathbb{R}^d\ \forall x \in \mathcal{X}, \qquad \mathrm{sgn}_\tau(\langle w, \phi_i^*(x)\rangle) = h(x). \tag{A.11}$$

Proof. Proposition A.5 gives $M \ge 1$, the displayed definition of the positive integer r, and $r \le 7TS$. Draw one random tuple $(\phi_1,\dots,\phi_r) \sim \mathcal{P}^r$ from the product of the single law fixed by Assumption 3. For each $h \in \mathcal{H}$, let

$$F_h := \bigcap_{i=1}^{r}\{\phi_i \in E_h^c\}, \tag{A.12}$$

where E_h is the exact event (A.8). Thus F_h is precisely the event that this tuple leaves h uncovered. By Lemma A.7,

$$\Pr_{\mathcal{P}^r}(F_h) \le 2^{-r} \qquad \text{for every } h \in \mathcal{H}. \tag{A.13}$$

The events F_h may be dependent because they are evaluated on the same tuple; no independence across targets is needed. Since $\mathcal{H}$ is finite, the pointwise indicator inequality

$$\mathbf{1}_{\cup_{h\in\mathcal{H}} F_h} \le \sum_{h\in\mathcal{H}} \mathbf{1}_{F_h}$$

and (A.13) imply

$$\Pr_{\mathcal{P}^r}\left[\bigcup_{h\in\mathcal{H}} F_h\right] \le \sum_{h\in\mathcal{H}} \Pr_{\mathcal{P}^r}(F_h) \le M2^{-r}. \tag{A.14}$$

The definition of r gives

$$r \ge \log_2(2M), \qquad M2^{-r} \le M2^{-\log_2(2M)} = \frac{1}{2} < 1. \tag{A.15}$$

Combining (A.14)– (A.15), the complementary simultaneous-coverage event

$$G := \bigcap_{h\in\mathcal{H}} F_h^c$$

satisfies

$$\Pr_{\mathcal{P}^r}(G) = 1 - \Pr_{\mathcal{P}^r}\left[\bigcup_{h\in\mathcal{H}} F_h\right] \geq \frac{1}{2} > 0. \tag{A.16}$$

The empty event has probability zero, so (A.16) proves that G contains at least one tuple. Fix one such realization and call it $(\phi_1^*, \ldots, \phi_r^*)$. This tuple is fixed once, before any target is selected from $\mathcal{H}$. For every $h \in \mathcal{H}$, membership in F_h^c gives some $i \in \{1, \ldots, r\}$ with $\phi_i^* \in E_h$; by the definition (A.8) of E_h, that membership gives some $w \in \mathbb{R}^d$ satisfying (A.11).

Thus the final quantifier order is

$$\exists(\phi_1^*, \ldots, \phi_r^*)\ \forall h \in \mathcal{H}\ \exists i \in \{1, \ldots, r\}\ \exists w \in \mathbb{R}^d\ \forall x \in \mathcal{X},$$

not a separate tuple or law chosen after h. No distribution appears in the fixed tuple or in (A.11), because the exact event (A.8) is independent of $\mathcal{D}$. □

Derivation of simultaneous deterministic coverage. Proposition A.5 supplies the positive integer

$$r = \lceil \log_2(2|\mathcal{H}|) \rceil \leq 7TS.$$

Assumption 3 supplies one law before all targets, and Lemma A.7 proves that r iid draws from this law miss any fixed target with probability at most 2^{-r}. Proposition A.6 pays the complete finite union through $|\mathcal{H}|2^{-r} \leq 1/2 < 1$ and fixes a tuple in the positive-probability simultaneous-coverage event. The resulting tuple is deterministic, common to all targets, full-domain, and tie-resolved exactly as in the premise. When $|\mathcal{H}| = 1$, the definition gives $r = 1$, and the construction reduces to the primitive success event for the sole target, whose probability is at least $1/2$. Thus success exactly at the stated threshold is sufficient. Any successful zero score remains valid because the same sgn_τ is used before and after fixing the tuple. □

A.5 Direct-sum representation and dimension closure

Proposition A.7 (Exact boundary routing and nondegenerate controls). *Under Assumption 1, Propositions A.1 and A.2, and Lemma A.1, the following exhaustive alternatives hold:*

1. *If $\mathcal{H} = \varnothing$, then $\operatorname{dc}(\mathcal{H}) = 0 \leq 7TSd$.*
2. *If $\mathcal{H} \neq \varnothing$ and $d = 0$, then $\operatorname{dc}(\mathcal{H}) = 0 = 7TSd$.*
3. *If $\mathcal{H} \neq \varnothing$ and $d \neq 0$, then $d \geq 1$, $S \geq n \geq 1$, and $T, S \geq 1$.*

Proof. The three cases are exhaustive because $\mathcal{H}$ is either empty or nonempty and $d \in \mathbb{Z}_{\geq 0}$. In the first case, Proposition A.1 gives the exact dimension-zero conclusion. In the second case, Lemma A.1 gives the exact common zero-map conclusion with the fixed convention $\operatorname{sgn}_\tau(0) = \tau$. In either case the logarithmic quantity r and a block tuple are unnecessary, so no undefined expression is introduced when $\mathcal{H} = \varnothing$. In the third case, Proposition A.2 supplies all displayed positivity and structural controls. □

Lemma A.8 (Exact direct-sum score preservation). *Under Assumption 1, Proposition A.2, and Proposition A.6, let $r = \lceil \log_2(2|\mathcal{H}|) \rceil$ and let $(\phi_1^*, \ldots, \phi_r^*)$ be the deterministic tuple supplied by Proposition A.6. If $\mathcal{H} \neq \varnothing$ and $d \neq 0$, then there are a single map $\Phi : \mathcal{X} \to \mathbb{R}^{rd}$ and vectors $u_h \in \mathbb{R}^{rd}$, one for each $h \in \mathcal{H}$, such that for every h and x,*

$$\langle u_h, \Phi(x) \rangle = \langle w_h, \phi_{i(h)}^*(x) \rangle, \qquad \operatorname{sgn}_\tau(\langle u_h, \Phi(x) \rangle) = h(x),$$

where $i(h)$ and w_h are one successful block and separator for h. The map Φ is independent of h.

Proof. By Proposition A.2, the branch has $d \geq 1$. Proposition A.6 supplies a fixed tuple and, for each h, a nonempty finite set

$$I_h := \left\{ i \in \{1, \ldots, r\} : \exists w \in \mathbb{R}^d \ \forall x \in \mathcal{X},\ \operatorname{sgn}_\tau(\langle w, \phi_i^*(x) \rangle) = h(x) \right\}.$$

Choose $i(h) := \min I_h$, and choose one witnessing vector $w_h \in \mathbb{R}^d$ for this index. This is a finite target-by-target selection of witnesses; it changes neither the already fixed tuple nor the feature map.

Define the concatenated map by

$$\Phi(x) := (\phi_1^*(x), \ldots, \phi_r^*(x)) \in (\mathbb{R}^d)^r \cong \mathbb{R}^{rd}.$$

This definition uses only the deterministic tuple and therefore is one common target-independent map. For each target define the zero-padded vector

$$u_h := (u_{h,1}, \ldots, u_{h,r}) \in (\mathbb{R}^d)^r, \qquad u_{h,j} := \begin{cases} w_h, & j = i(h), \\ 0_d, & j \neq i(h), \end{cases}$$

where 0_d is the zero vector in $\mathbb{R}^d$. The Euclidean product inner product gives, for every $x \in \mathcal{X}$,

$$\begin{aligned} \langle u_h, \Phi(x) \rangle &= \sum_{j=1}^{r} \langle u_{h,j}, \phi_j^*(x) \rangle \\ &= \langle w_h, \phi_{i(h)}^*(x) \rangle, \end{aligned}$$

because every term with $j \neq i(h)$ is $\langle 0_d, \phi_j^*(x) \rangle = 0$. Thus the score residual is explicitly

$$\langle u_h, \Phi(x) \rangle - \langle w_h, \phi_{i(h)}^*(x) \rangle = 0.$$

Applying the same fixed sgn_τ to equal scores preserves the successful block's exact label, including when that score is zero. Hence $\operatorname{sgn}_\tau(\langle u_h, \Phi(x) \rangle) = h(x)$ for every $h \in \mathcal{H}$ and $x \in \mathcal{X}$. ☐

Proposition A.8 (Common exact representation and dimension bound)**.** *Under Assumption 1, Propositions A.2, A.5, and A.6, and Lemma A.8, if $\mathcal{H} \neq \varnothing$ and $d \neq 0$, then*

$$\operatorname{dc}(\mathcal{H}) \leq rd \leq 7TSd, \qquad r = \lceil \log_2(2|\mathcal{H}|) \rceil.$$

Proof. Lemma A.8 supplies one map $\Phi : \mathcal{X} \to \mathbb{R}^{rd}$ and, for every target, a vector $u_h \in \mathbb{R}^{rd}$ with exact tie-resolved signs on all of $\mathcal{X}$. By the definition of dc, this is a feasible representation of dimension rd, so $\operatorname{dc}(\mathcal{H}) \leq rd$. Proposition A.5 gives $r \leq 7TS$. Since $d \geq 1$ by Proposition A.2,

$$7TSd - rd = (7TS - r)d \geq 0,$$

and therefore $rd \leq 7TSd$. The integer product rd is a valid dimension because $r, d \in \mathbb{Z}_{\geq 0}$. ☐

Corollary A.1 (Conditional polynomial rate specialization)**.** *Under Assumption 1, Proposition A.7, and Proposition A.8 on the nondegenerate branch, suppose additionally that a separate result supplies an explicit polynomial expression $p(S,T)$ satisfying $d \leq p(S,T)$ at the current setup, with all coefficients and auxiliary dependence exposed and with no hidden dependence on n or η. Then every branch satisfies*

$$\operatorname{dc}(\mathcal{H}) \leq 7TSp(S,T),$$

which is polynomial in S, T whenever p is polynomial in those variables.

Proof. The additional inequality is a local conditional hypothesis for this corollary only; it is not used to establish Proposition A.8. From Assumption 1, $T \geq 1$, and the positive architecture widths imply $S \geq 1$; hence $7TS \geq 0$. The supplied inequality and $d \geq 0$ imply $p(S,T) \geq d \geq 0$ at the current parameters. Propositions A.7 and A.8, applied in the nondegenerate alternative, jointly give the all-branch main inequality

$$\mathrm{dc}(\mathcal{H}) \leq 7TSd.$$

Indeed, the first proposition closes $\mathcal{H} = \varnothing$ and $d = 0$, and the second closes the remaining branch. The exact absorption relation is

$$7TSp(S,T) - 7TSd = 7TS\big(p(S,T) - d\big) \geq 0.$$

Combining the all-branch main inequality with this relation gives

$$\mathrm{dc}(\mathcal{H}) \leq 7TSd \leq 7TSp(S,T).$$

No representation score, tie, probability, or horizon term is changed: on the nondegenerate branch the same deterministic map from Lemma A.8 is used, and on the null branches the exact maps remain in force. Only a scalar upper bound is substituted. If the separate inequality is unavailable, this corollary is not asserted. When $p(S,T) = d$, the bridge is equality; when $d = 0$, the exact boundary conclusion of Lemma A.1 remains the baseline rather than being replaced by a positive-dimensional surrogate. □

Derivation of the dimension bound. Proposition A.7 first handles every boundary branch. If $\mathcal{H} = \varnothing$, the exact convention gives $\mathrm{dc}(\mathcal{H}) = 0$, and no logarithmic budget or feature tuple is needed. If $\mathcal{H} \neq \varnothing$ and $d = 0$, Lemma A.1 gives $\mathrm{dc}(\mathcal{H}) = 0 = 7TSd$ with the original tie label.

On the remaining branch, Proposition A.2 supplies $d \geq 1$ and $T, S \geq 1$. Proposition A.5 supplies the exact integer $r = \lceil \log_2(2|\mathcal{H}|) \rceil \leq 7TS$, while Proposition A.6 supplies one deterministic tuple that works for every target. Lemma A.8 concatenates that tuple into one map Φ chosen before any target-specific separator. Its pointwise score identity is exact, so the fixed sgn_τ, including zero-score ties, returns each target exactly. Proposition A.8 then invokes the definition of dc and the explicit comparison $rd \leq 7TSd$. Thus every setup under the three assumptions obeys

$$\mathrm{dc}(\mathcal{H}) \leq 7TSd.$$

The common map is target-independent: the tuple is fixed once, whereas only $i(h)$, w_h, and u_h depend on h, exactly as permitted in the definition of dc. Corollary A.1 performs the optional specialization only when a separate explicit inequality $d \leq p(S,T)$ is available. When $r = 1$, the direct sum is the successful block itself. When $p(S,T) = d$, the polynomial comparison is equality, and when $d = 0$ the exact zero-dimensional conclusion is retained instead of replaced by a positive surrogate. □

A.6 Proof of the Main Theorem

Proof of Theorem 3.1. Under Assumptions 1, 2, and 3, Proposition A.7 gives an exhaustive case split. Its first two alternatives establish the result directly when $\mathcal{H} = \varnothing$ or $d = 0$. In its third alternative, Proposition A.8, whose inputs are supplied by Propositions A.4, A.5, and A.6, gives

$$\mathrm{dc}(\mathcal{H}) \leq rd \leq 7TSd.$$

Every branch therefore satisfies the claimed deterministic bound. The only use of Assumption 2 is the VC ceiling in Proposition A.4; the target-independent confident-map premise remains a separate explicit condition and is not derived from SGD. □